Index laws

1

a 2^{21}	b 22^{8}	c 11^{4}	d 7^{5}
e 5^{48}	f 15^{4}	g 23^{79}	h 17^{8}
i 19^{5}	j 12^{7}	k 13^{8}	l 15^{27}

2

a 8	b 7	c 34	d 5
e 3	f 6	g 5	h 38
i 5	j 2		

3

a 1 679 616	b 9 261	c 1 024	d 390 625
e 16 807	f 4 096	g 121	h 729
i 36	j 1	k 1 728	l 36

Negative bases

4

a $(-11)^{13}$	b $(-21)^{8}$	c -15^{2}	d $(-3)^{7}$
e -12	f $(-30)^{3}$	g 8^{3}	h 11^{11}
i $(-7)^{5}$	j $(-6)^{2}$		

5

a -7	b -3	c 50	d 2
e 49	f 12	g -2	h -21
i 3	j -55		

6

a 256	b 64	c -243	d $-8\,000$
e -27	f $-3\,125$	g -216	h $-1\,977\,326\,743$
i 1 225	j 531 441	k 64	l $-8\,000$

Fractional bases

7

a $\frac{1}{3^4}$ b $\frac{3^3}{8^3}$ c $\frac{1^8}{4^8}$ d $\frac{5^2}{2^2}$

e $\frac{10^5}{33^5}$ f $\frac{2^6}{35^6}$ g $\frac{5^3}{18^3}$ h $\frac{29^7}{41^7}$

i $\frac{11^9}{13^9}$ j $\frac{20^2}{3^2}$ k $\frac{17^4}{4^4}$ l $\frac{31^5}{50^5}$

8

a 3 b 3 c 2 d 1

e 3 f 16 g 9 h 625

i 2 j 121 k 5 l 5

9

a $\frac{1}{9}$ b $\frac{27}{125}$ c $\frac{1}{32}$ d $\frac{49}{121}$

e $\frac{27}{512}$ f $\frac{144}{169}$ g $\frac{125}{729}$ h $\frac{16}{81}$

i $\frac{4\,096}{15\,625}$ j $\frac{8}{125}$ k $\frac{25}{49}$ l $\frac{1\,000}{2\,744}$

10

a No b Yes c No d Yes

Negative indices

11 Less than 1.

12

a $\frac{1}{a^9}$ b a^n c $\frac{1}{4a^9}$ d $\frac{b^m}{a^n}$

e $\frac{1}{p^2}$ f $\frac{3}{x^4}$ g $\frac{7}{x^9}$ h $\frac{q^3}{p^2}$

i $\frac{8}{p^3}$ j $\frac{2y^3}{x^8}$ k $\frac{c^3}{6b^7}$ l $\frac{p^4}{m^5n^4}$

m $\frac{3a^4}{b^5c^7}$ n $\frac{l^7}{16k^3}$ o $\frac{g^2}{f^7h^8}$ p $\frac{s^4u^6}{5t^9}$

13

a y^{-4} b $2x^{-6}$ c xy^{-2} d $4p^{-6}q^{-7}$

14

a $20y^6$ b $28a^2$ c $\frac{-15}{x^4}$ d $\frac{12}{y^5}$

e $8h^7$ f $\frac{6}{y^7}$ g $\frac{16}{y^3}$ h $25m^3$

i $\frac{3}{x^7}$ j $\frac{3}{x^4}$ k $3x^8$ l $\frac{3}{2c^2}$

m $\frac{5h^9}{4}$ n $3x^8$ o $\frac{3}{2p}$ p $\frac{9}{4w^{10}}$

15

a $\frac{1}{8m^3}$ b $\frac{256}{m^{24}}$ c $\frac{p^8}{9}$ d $\frac{1}{9y^4}$

e $\frac{64}{y^3}$ f $\frac{y^4}{x^5}$ g $\frac{y^{36}}{x^{28}}$ h $5x^5$

i $\frac{5}{x^4}$ j $\frac{81}{z^4}$ k $\frac{q^7}{p^3}$ l $\frac{x^8}{y^{16}}$

16

a $n = -2$ b $n = -3$ c $n = -4$ d $n = -3$

17 $\left(x^4y^{-3}\right)^{-4} = \frac{y^{12}}{x^{16}}$

18 The -1 index only applies to the term directly in front of it. In this case it only applies to the a, not the 5. The correct working is $5a^{-1} = \frac{5}{a}$.

19

a $32y^8$ b $60y^5$

20

a $\frac{10z^9}{21x^3y^7}$ b $\frac{160y^6}{27x}$